One Field: Gravity, Dark Matter, Dark Energy, and the Arrow of Time in Event Density

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Status, in one honest paragraph

Everything below is a **conditional structural result**: conditional on thirteen postulated substrate primitives, on two scales taken from observation — the Planck scale and the cosmic expansion rate, with c and $\hbar$ — on one labeled postulate (the linearity of the commitment rate), and on one matching to a previously derived law. Newton's constant is *not* a third input: the program derives its form, $G = c^3\ell_P^2/\hbar$ (Paper 027 / T19), so G and the Planck scale are *one* inherited scale, not two — G is a downstream coefficient, not a fundamental constant. Nothing below is claimed as confirmed physics. Every condition, inheritance, and open front is itemized in the six component papers, each of which opens with what it does *not* claim. What this letter asserts is narrower and stronger than a confirmation: **a falsifiable unification** — four sectors of gravitational physics carried by one field with one scale, with named tests and, in one case, a specific number to miss.

The claim

> **From a discrete substrate whose only special structure is an irreversible arrow of time, one derives: Einstein's weak-field metric, including the factor-of-two light bending; the precise gravitational theory ED must be (khronometric — Einstein's gravity plus exactly one scalar, the arrow itself made dynamical); the MOND phenomenology of galaxies, with its transition scale $a_0 = cH_0/2\pi$ carried structurally rather than tuned; and the dark-energy scale, as the same field's vacuum term. One field. One scale. Five roles: the arrow, the Einstein sector, the dark-matter sector, the cosmological clustering dial, and the cosmological constant's magnitude.**

The rest of this letter is that sentence, unpacked, with its supports and its kill-switches.

The chain, in five steps

1. The substrate and the arrow. Event Density is a discrete relational substrate — thirteen primitives, no fundamental spacetime — whose defining commitment is that becoming is *irreversible*: facts are committed and never uncommitted, at a homogeneous tick. Where standard physics puts the arrow of time in boundary conditions atop time-symmetric laws, ED puts it **in the law**. Everything in this letter flows from that one inversion.

2. The metric, and the factor of two *(Paper GR-I).* The substrate's bandwidth field b — its capacity for connection — supplies an emergent spatial metric, $g \sim 1/b$. The new result is the *temporal* lapse: requiring the substrate's fastest signal to define the null cone, together with the commitment-rate law, forces $N^2 \sim b$ — the **same field sets how space curves and how fast clocks run**. That locking yields the Schwarzschild relation $g_{00} \cdot g_{rr} = -1$, and with it the three classical tests: gravitational redshift, perihelion precession, and the famous **factor-of-two light bending** (ray-traced in simulation: ratio 2.09 against the predicted 2.00, with the no-bending conformal control at exactly zero). The factor of two — historically the discriminator that killed scalar gravity and crowned Einstein's — emerges here because one substrate field cannot help but curve space and time together.

3. The class: the arrow is the deviation *(Paper GR-II).* Which gravity is this? Lovelock's uniqueness theorem forces the field equation toward Einstein's — *unless* an extra field propagates. One does, and its identity is the theory's signature: because the arrow lives in the law, the substrate carries a preferred time-foliation, and the propagating modes are Einstein's two tensor polarizations **plus one scalar — the khronon, the arrow made dynamical**. ED's gravity is khronometric (the infrared limit of Hořava gravity): *Einstein plus the arrow, and nothing else.* The same unification that produces the scalar protects the theory where preferred-frame gravities usually die: matter and geometry are one substrate with **one causal cone**, so the gravitational-wave speed equals the light speed *structurally* (the post-GW170817 constraint as an identity, not a tuning), and Lorentz violation is *universal* rather than species-dependent — invisible to the 10^{-20} matter-sector bounds that execute the generic theories of this class.

4. Dark matter: the arrow's deep field *(Paper KM-I).* ED's earlier gravity papers had separately derived the MOND phenomenology — the transition acceleration $a_0 = cH_0/2\pi$ from the cosmic horizon, and the deep-field law $a = \sqrt{(a_N \cdot a_0)}$ that fits galactic rotation without dark matter. KM-I supplies the missing identification: **that sector is the khronon's deep-infrared regime.** The construction's only hinge is matching the already-derived deep-field law, which fixes the khronon's kinetic term uniquely; the scale a_0 needs no tuning because the khronon's background *is* cosmic time — the long-noticed Milgrom coincidence $a_0 \approx cH_0/2\pi$ acquires a structural carrier. And the historical kill-checks of relativistic MOND pass *structurally*: galactic lensing tracks the MOND potential **with no added vector field** (the modification is metric-borne; the conformal-coupling disease TeVeS was invented to patch never arises), no ghost appears (the MOND sector adds no time derivatives), and the solar system is screened. *Dark matter, in this picture, is not a particle. It is the deep regime of the field that makes time irreversible.*

5. The cosmological face: the dial, and one Λ *(Paper KM-II).* Clusters and the CMB are where MOND-class theories die: the CMB wants something that clusters like cold dust. The khronon carries a cosmological sector positioned exactly there — not added for the purpose, but forced by the effective theory's own generality — and **provably sequestered** from every galactic test (the Orthogonality Theorem: the sector vanishes identically in static situations, so it can neither retrofit the galaxy results nor hide behind them). On cosmological backgrounds it is a dark fluid, and the near-zero sound speed that looked like the theory's one soft spot is *precisely* the dust-clustering condition the CMB demands — **the vulnerability and the opportunity are the same dial**, constrained by five filters. Finally, Λ: the khronon's vacuum term and ED's earlier substrate-level Λ mechanism carry the same scale, $H_0^2 c^2/G$, because both are projections of one cosmological boundary — **one Λ at two levels of description**, with the matching performed: the scaling exact, the sign check passed, and the effective theory's vacuum constant pinned at $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$ — leaving the underlying substrate integral as an open computation **with a specific number it must hit**.

Why this is not trivial

Four scales that have no business coinciding — the rate of the cosmic arrow, the acceleration where galactic dynamics turn non-Newtonian, the energy density of the vacuum, and the locus where horizons form — are carried here by **one scalar field with one background scale**. In any framework where these are independent sectors, their numerical coincidences ($a_0 \approx cH_0/2\pi$; $\rho_\Lambda \sim H_0^2 c^2/G$) are unexplained accidents. Here they are the same number wearing four hats, because the field has only one number to offer. And the field itself was not invented for any of these jobs: it was *counted* — forced into existence by a mode-counting argument about the arrow of time — before dark matter or dark energy were on the table. The theory spends a field it already had.

It is also worth saying what kind of construction this is *not*. No vector field is added (TeVeS's patch is unneeded — the disease never arises). No interpolation function is chosen (the two asymptotic regimes are fixed by inheritance and matching; the middle is an honest, observationally constrained family). No parameter is tuned to GW170817 (the constraint is an identity of the single-cone structure). The deviations from textbook General Relativity are not scattered: they are located, all of them, at the one place ED differs from standard physics by construction — the arrow.

What would kill it

This unification is falsifiable on three independent fronts, and says so:

1. **The preferred-frame front.** The khronon's coupling parameters (α_1, α_2 in the PPN expansion) must sit inside the solar-system and pulsar bounds or the theory dies. This has been the declared falsification target since GR-II. **(Note added, GR-IV — *The Arrow's Alibi.*)** Both are now resolved. The *tighter* parameter vanishes outright: $\alpha_2 = 0$ exactly, a structural consequence of both gravitational cones being at light speed ($c_T = c$, $c_s = c$), literature-verified — so the tighter bound is met for free. The remaining one is the *local* khronon stiffness, $\alpha_1 = -4\lambda_{local}$ with $\lambda_{local} \sim \rho_{event}/\rho_{Planck}$, suppressed by the sparseness of commitment — itself *forced* by ED's quantum sector (dense commitment is the quantum Zeno limit, and would abolish quantum mechanics). This puts α_1 below the

bound by $\geq$70 orders, so the front is closed *favorably.* It remains live as a **test**: a measured $\alpha_1 \gtrsim 10^{-5}$, or *any* nonzero α_2, in any sub-Planck environment still kills the theory.

2. **The galactic front.** The deep-field law with the *inherited* a_0 must keep fitting: wide-binary stars near a_0 are a live observational discriminator right now; a clean Newtonian result at the relevant precision kills the unscreened sector. The lensing identity (no slip beyond $O(\Phi)$) is likewise checkable, and there is no vector field to repair it with.
3. **The cosmological front.** The dial member that survives the five filters must deliver dust-like clustering of the right abundance at recombination — the CMB is the judge, and by the Orthogonality Theorem the galactic sector can neither help nor be blamed. And the one-Λ thesis now carries a *number*: the ab-initio substrate evaluation of the boundary integral must land on $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$. Wrong sign or wrong magnitude breaks the thesis. **A theory with a number to miss is a theory that can lose.**

What is claimed, and what is not

Claimed: a conditional, falsifiable, structurally economical unification — the weak-field Einstein metric with its classical tests; the khronometric class with the arrow as its single deviation from GR; the MOND sector as that scalar's deep regime, lensing-correct without additions; the cosmological sector sequestered, dust-capable, and carrying the Λ-scale — all from one field whose origin (the arrow) predates all of its jobs.

Not claimed: that GR is "cracked" or re-derived from nothing (the Planck scale ℓ_P and the cosmic rate H_0 are the inherited scales; $G = c^3\ell_P^2/\hbar$ and $a_0 = cH_0/2\pi$ are *derived coefficients* of those scales, not independent inputs; one postulate and one matching are labeled and load-bearing); that dark matter or dark energy are "solved" (the clusters/CMB debt is *addressable, not discharged*; no Λ value is derived — the matching transports the inherited tier, it does not upgrade it); that the matter-sector cosmological-constant problem is touched; or that any of this is confirmed. The five-filter dial is unset. *(The preferred-frame numbers, formerly uncomputed, are computed and suppressed in GR-IV — see the kill-switch note above.)* The deep question — *why* the substrate's response goes geometric-mean below the cosmic rate — is deliberately left open rather than reverse-engineered.

Where the proofs live

Step	Paper	What it carries
The metric and the factor of two	**GR-I** — *The Bandwidth Lapse and the Factor of Two*	the lapse derivation; the Schwarzschild relation; the three classical tests; the ray-tracing simulation
The class and the arrow	**GR-II** — *The Arrow's Fingerprint*	Lovelock; the mode count; $c_T = c$; universal Lorentz violation; the α_1, α_2 front

Step	Paper	What it carries
The dynamical rule and the weak-field closure	**GR-III** — *The Arrow's Engine*	the commitment-rate law; the dynamical field equation; the closure of GR-I's labeled weak-field assumptions
Dark matter as the deep field	**KM-I** — *The Arrow's Deep Field*	the kinetic matching; the lensing pass; screening; the Bandwidth–Acceleration Lemma
The cosmological face and Λ	**KM-II** — *The Arrow's Horizon*	the Orthogonality Theorem; the clustering dial; the one-Λ matching and the pinned $\mathcal{W}_0$
The preferred-frame front, closed	**GR-IV** — *The Arrow's Alibi*	$\alpha_1 = -4\lambda_{local}$; $\lambda_{local} \sim \rho_{event}/\rho_{Planck}$; sparse commitment forced by the quantum sector (Zeno); α_1 safe by $\geq$70 orders; MOND on the cosmic boundary

Each component paper opens with its full does-not-claim preamble and carries a step-by-step load-bearing audit; the underlying working record (nineteen foundations rounds) is preserved in the program's working repository.

Closing

The ordinary reading of physics puts time-symmetry at the bottom and lets the arrow emerge. Event Density inverts that — the arrow goes in the law — and this letter reports what fell out when that inversion was pushed through gravity without flinching: Einstein's weak field, the one permitted deviation, the galaxies, and the vacuum, hanging from a single scalar anchored to a single scale. It may be wrong. It says exactly where to look to find out. That is the most a theory at this stage can honestly offer — and it is more than most offer.

One field. One scale. Five roles. Three ways to kill it. One number to miss.

End of Letter.

Capstone letter for the post-pivot gravity line (GR-I, GR-II, GR-III, GR-IV, KM-I, KM-II). All derivations, audits, inheritances, postulates, and open fronts live in the component papers; this letter adds no new result and softens no stated limit.